

```
import os
os.environ["OPENAI_API_KEY"] = "sk-proj-"
```

```
!pip install langchain chromadb openai tiktoken pypdf langchain_openai langchain-community
```

```
Downloading onnxruntime-1.21.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (16.0 MB)
16.0/16.0 MB 62.6 MB/s eta 0:00:00
Downloading opentelemetry_exporter_otlp_proto_grpc-1.31.1-py3-none-any.whl (18 kB)
Downloading opentelemetry_exporter_otlp_proto_common-1.31.1-py3-none-any.whl (18 kB)
Downloading opentelemetry_proto-1.31.1-py3-none-any.whl (55 kB)
55.9/55.9 kB 4.4 MB/s eta 0:00:00
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Downloading opentelemetry_instrumentation_asgi-0.52b1-py3-none-any.whl (16 kB)
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Downloading overrides-7.7.0-py3-none-any.whl (17 kB)
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Downloading durationpy-0.9-py3-none-any.whl (3.5 kB)
Downloading httptools-0.6.4-cp311-cp311-manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_17_x86_64
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Downloading pyproject_hooks-1.2.0-py3-none-any.whl (10 kB)
Downloading asgiref-3.8.1-py3-none-any.whl (23 kB)
Downloading humanfriendly-10.0-py2.py3-none-any.whl (86 kB)
86.8/86.8 kB 7.0 MB/s eta 0:00:00
Downloading mpy_extensions-1.0.0-py3-none-any.whl (4.7 kB)
Building wheels for collected packages: pypika
  Building wheel for pypika (pyproject.toml) ... done
  Created wheel for pypika: filename=pypika-0.48.9-py2.py3-none-any.whl size=53800 sha256=24b2a79e17
  Stored in directory: /root/.cache/pip/wheels/a3/01/bd/4c40ceb9d5354160cb186dcc153360f4ab7eb23e2b24
Successfully built pypika
Installing collected packages: pypika, monotonic, durationpy, uvloop, uvicorn, python-dotenv, pyproj
  Attempting uninstall: langchain-core
    Found existing installation: langchain-core 0.3.50
    Uninstalling langchain-core-0.3.50:
      Successfully uninstalled langchain-core-0.3.50
  Attempting uninstall: langchain-text-splitters
    Found existing installation: langchain-text-splitters 0.3.7
    Uninstalling langchain-text-splitters-0.3.7:
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  Attempting uninstall: langchain
    Found existing installation: langchain 0.3.22
    Uninstalling langchain-0.3.22:
      Successfully uninstalled langchain-0.3.22
Successfully installed asgiref-3.8.1 backoff-2.2.1 bcrypt-4.3.0 build-1.2.2.post1 chroma-hnswlib-0.7
```

```
from langchain_openai import OpenAIEmbeddings
from langchain.vectorstores import Chroma
```

```
from langchain.schema import Document

# Create LangChain documents for IPL players

doc1 = Document(
    page_content="Virat Kohli is one of the most successful and consistent batsmen in IPL history"
    metadata={"team": "Royal Challengers Bangalore"}
)
doc2 = Document(
```

```

        page_content="Rohit Sharma is the most successful captain in IPL history, leading Mumbai Ind
        metadata={"team": "Mumbai Indians"}
    )
doc3 = Document(
    page_content="MS Dhoni, famously known as Captain Cool, has led Chennai Super Kings to multi
    metadata={"team": "Chennai Super Kings"}
)
doc4 = Document(
    page_content="Jasprit Bumrah is considered one of the best fast bowlers in T20 cricket. Play
    metadata={"team": "Mumbai Indians"}
)
doc5 = Document(
    page_content="Ravindra Jadeja is a dynamic all-rounder who contributes with both bat and bal
    metadata={"team": "Chennai Super Kings"}
)

```

```
docs = [doc1, doc2, doc3, doc4, doc5]
```

```

vector_store = Chroma(
    embedding_function=OpenAIEmbeddings(),
    persist_directory='my_chroma_db',
    collection_name='sample'
)

```

```

# add documents
vector_store.add_documents(docs)

```

```

['09a39dc6-3ba6-4ea7-927e-fdda591da5e4',
 '8b561bf2-72ce-4295-8097-27e6f3bcd582',
 'aa800a1a-4b4f-4e58-8fce-ae2279eb385b',
 'eb519603-4222-46d5-a596-11797a4d39b6',
 '1182b187-1e5b-4e2d-a076-0119742940cc']

```

```

# view documents
vector_store.get(include=['embeddings','documents', 'metadatas'])

```

```

{'ids': ['09a39dc6-3ba6-4ea7-927e-fdda591da5e4',
 '8b561bf2-72ce-4295-8097-27e6f3bcd582',
 'aa800a1a-4b4f-4e58-8fce-ae2279eb385b',
 'eb519603-4222-46d5-a596-11797a4d39b6',
 '1182b187-1e5b-4e2d-a076-0119742940cc'],
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                        -0.00366616,  0.01357884],
                       [ -0.00268021, -0.00010323,  0.02815653, ..., -0.01501936,
                        0.00590092, -0.01164922],
                       [  0.00092799, -0.00476     ,  0.0124662 , ..., -0.01731381,
                        0.00075886,  0.00296567],
                       [ -0.02714536,  0.00885395,  0.02699314, ..., -0.02592762,
                        0.00900617, -0.01999116],
                       [ -0.01810451,  0.01281202,  0.0347942 , ..., -0.03034012,
                        -0.00595078,  0.00521716]]),
 'documents': ['Virat Kohli is one of the most successful and consistent batsmen in IPL history.
Known for his aggressive batting style and fitness, he has led the Royal Challengers Bangalore in
multiple seasons.',
 "Rohit Sharma is the most successful captain in IPL history, leading Mumbai Indians to five titles.
He's known for his calm demeanor and ability to play big innings under pressure.",
 'MS Dhoni, famously known as Captain Cool, has led Chennai Super Kings to multiple IPL titles. His
finishing skills, wicketkeeping, and leadership are legendary.',
 'Jasprit Bumrah is considered one of the best fast bowlers in T20 cricket. Playing for Mumbai
Indians, he is known for his yorkers and death-over expertise.',
 'Ravindra Jadeja is a dynamic all-rounder who contributes with both bat and ball. Representing
Chennai Super Kings, his quick fielding and match-winning performances make him a key player.'],
 'uris': None,
 'included': ['embeddings', 'documents', 'metadatas'],
 'data': None,
 'metadatas': [{'team': 'Royal Challengers Bangalore'},
 {'team': 'Mumbai Indians'},
 {'team': 'Chennai Super Kings'},
 {'team': 'Mumbai Indians'},
 {'team': 'Chennai Super Kings'}]}

```

```

# search documents
vector_store.similarity_search(
    query='Who among these are a bowler?',

```

```

    k=2
)

[Document(metadata={'team': 'Mumbai Indians'}, page_content='Jasprit Bumrah is considered one of the best fast bowlers in T20 cricket. Playing for Mumbai Indians, he is known for his yorkers and death-over expertise.'),
 Document(metadata={'team': 'Chennai Super Kings'}, page_content='Ravindra Jadeja is a dynamic all-rounder who contributes with both bat and ball. Representing Chennai Super Kings, his quick fielding and match-winning performances make him a key player.')]

```

```

# search with similarity score
vector_store.similarity_search_with_score(
    query='Who among these are a bowler?',
    k=2
)

```

```

[(Document(metadata={'team': 'Mumbai Indians'}, page_content='Jasprit Bumrah is considered one of the best fast bowlers in T20 cricket. Playing for Mumbai Indians, he is known for his yorkers and death-over expertise.'),
 0.35445845127105713),
 (Document(metadata={'team': 'Chennai Super Kings'}, page_content='Ravindra Jadeja is a dynamic all-rounder who contributes with both bat and ball. Representing Chennai Super Kings, his quick fielding and match-winning performances make him a key player.'),
 0.4085893929004669)]

```

```

# meta-data filtering
vector_store.similarity_search_with_score(
    query="",
    filter={"team": "Chennai Super Kings"}
)

```

```

[(Document(metadata={'team': 'Chennai Super Kings'}, page_content='MS Dhoni, famously known as Captain Cool, has led Chennai Super Kings to multiple IPL titles. His finishing skills, wicketkeeping, and leadership are legendary.'),
 0.6488258242607117),
 (Document(metadata={'team': 'Chennai Super Kings'}, page_content='Ravindra Jadeja is a dynamic all-rounder who contributes with both bat and ball. Representing Chennai Super Kings, his quick fielding and match-winning performances make him a key player.'),
 0.6566494703292847)]

```

```

# update documents
updated_doc1 = Document(
    page_content="Virat Kohli, the former captain of Royal Challengers Bangalore (RCB), is renowned
    metadata={"team": "Royal Challengers Bangalore"}
)

vector_store.update_document(document_id='09a39dc6-3ba6-4ea7-927e-fdda591da5e4', document=updated_do

```

```

# view documents
vector_store.get(include=['embeddings', 'documents', 'metadatas'])

```

```

{'ids': ['09a39dc6-3ba6-4ea7-927e-fdda591da5e4',
 '8b561bf2-72ce-4295-8097-27e6f3bcd582',
 'aa800a1a-4b4f-4e58-8fce-ae2279eb385b',
 'eb519603-4222-46d5-a596-11797a4d39b6',
 '1182b187-1e5b-4e2d-a076-0119742940cc'],
 'embeddings': array([[ -0.00544442, -0.01907989,  0.00706373, ..., -0.01627786,
    -0.00032134,  0.00724619],
 [ -0.00268021, -0.00010323,  0.02815653, ..., -0.01501936,
    0.00590092, -0.01164922],
 [ 0.00092799, -0.00476 ,  0.0124662 , ..., -0.01731381,
    0.00075886,  0.00296567],
 [ -0.02714536,  0.00885395,  0.02699314, ..., -0.02592762,
    0.00900617, -0.01999116],
 [ -0.01810451,  0.01281202,  0.0347942 , ..., -0.03034012,
    -0.00595078,  0.00521716]]),
 'documents': ["Virat Kohli, the former captain of Royal Challengers Bangalore (RCB), is renowned for his aggressive leadership and consistent batting performances. He holds the record for the most runs in IPL history, including multiple centuries in a single season. Despite RCB not winning an IPL title under his captaincy, Kohli's passion and fitness set a benchmark for the league. His ability to chase targets and anchor innings has made him one of the most dependable players in T20 cricket.",
 "Rohit Sharma is the most successful captain in IPL history, leading Mumbai Indians to five titles. He's known for his calm demeanor and ability to play big innings under pressure.",
 "MS Dhoni, famously known as Captain Cool, has led Chennai Super Kings to multiple IPL titles. His finishing skills, wicketkeeping, and leadership are legendary.",
 "Jasprit Bumrah is considered one of the best fast bowlers in T20 cricket. Playing for Mumbai

```

```
Indians, he is known for his yorkers and death-over expertise.',
'Ravindra Jadeja is a dynamic all-rounder who contributes with both bat and ball. Representing
Chennai Super Kings, his quick fielding and match-winning performances make him a key player.'],
'uris': None,
'included': ['embeddings', 'documents', 'metadatas'],
'data': None,
'metadatas': [{'team': 'Royal Challengers Bangalore'},
{'team': 'Mumbai Indians'},
{'team': 'Chennai Super Kings'},
{'team': 'Mumbai Indians'},
{'team': 'Chennai Super Kings'}]}
```

```
# delete document
vector_store.delete(ids=['09a39dc6-3ba6-4ea7-927e-fdda591da5e4'])
```

```
# view documents
vector_store.get(include=['embeddings', 'documents', 'metadatas'])
```

```
{'ids': ['8b561bf2-72ce-4295-8097-27e6f3bcd582',
'aa800a1a-4b4f-4e58-8fce-ae2279eb385b',
'eb519603-4222-46d5-a596-11797a4d39b6',
'1182b187-1e5b-4e2d-a076-0119742940cc'],
'embeddings': array([[ -0.00268021, -0.00010323,  0.02815653, ..., -0.01501936,
  0.00590092, -0.01164922],
 [ 0.00092799, -0.00476   ,  0.0124662 , ..., -0.01731381,
  0.00075886,  0.00296567],
 [-0.02714536,  0.00885395,  0.02699314, ..., -0.02592762,
  0.00900617, -0.01999116],
 [-0.01810451,  0.01281202,  0.0347942 , ..., -0.03034012,
 -0.00595078,  0.00521716]]),
'documents': ["Rohit Sharma is the most successful captain in IPL history, leading Mumbai Indians to
five titles. He's known for his calm demeanor and ability to play big innings under pressure.",
'MS Dhoni, famously known as Captain Cool, has led Chennai Super Kings to multiple IPL titles. His
finishing skills, wicketkeeping, and leadership are legendary.',
'Jasprit Bumrah is considered one of the best fast bowlers in T20 cricket. Playing for Mumbai
Indians, he is known for his yorkers and death-over expertise.',
'Ravindra Jadeja is a dynamic all-rounder who contributes with both bat and ball. Representing
Chennai Super Kings, his quick fielding and match-winning performances make him a key player.'],
'uris': None,
'included': ['embeddings', 'documents', 'metadatas'],
'data': None,
'metadatas': [{'team': 'Mumbai Indians'},
{'team': 'Chennai Super Kings'},
{'team': 'Mumbai Indians'},
{'team': 'Chennai Super Kings'}]}
```

Start coding or [generate](#) with AI.